

PowerShell Downloader Activity Detection and Investigation using Sysmon and Wazuh

Objective

The objective of this simulation was to investigate how Sysmon and Wazuh detect PowerShell-based downloader behavior involving DNS queries, outbound HTTP communication, and suspicious PowerShell network activity commonly associated with malware staging and payload delivery.

MITRE ATTACK Mapping

Tactic	Technique	ID
Execution	PowerShell	T1059.001
Command and Control	Application Layer Protocol	T1071
Discovery	System Network Configuration Discovery	T1016

Attack Simulation

The following commands were executed in the Windows endpoint lab environment:

```
powershell.exe -ep bypass -Command "Invoke-WebRequest http://example.com -Out File test.html; notepad.exe test.html"
```

Attack Chain Overview

Step	Activity
1	PowerShell launched with execution policy bypass
2	DNS query performed
3	Outbound HTTP connection established
4	File downloaded
5	Downloaded file opened using Notepad

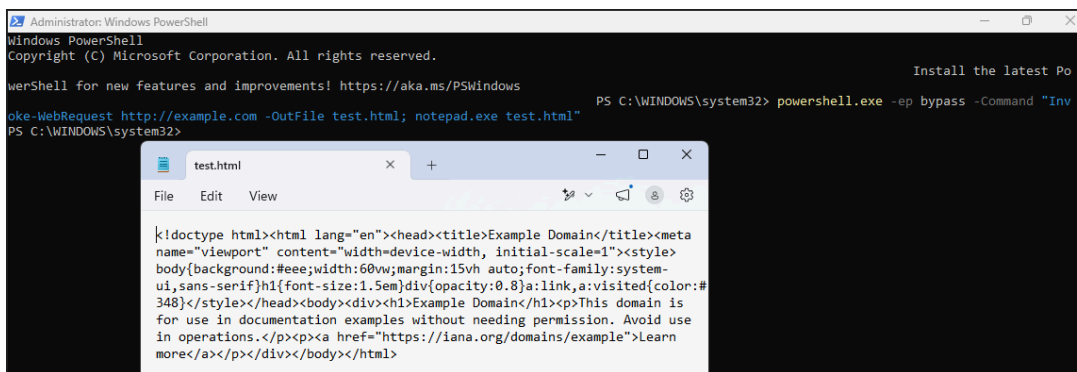


Figure 1 — PowerShell downloader activity executed in the lab environment

Sysmon Investigation

Key Sysmon Events Identified

Sysmon Event ID	Detection Purpose
1	Process Creation
3	Network Connection
22	DNS Query
11	File Creation

These events formed the primary telemetry chain used during the investigation.

Sysmon Event ID 1 — Process Creation Analysis

Key Findings

Field	Value
Event ID	1
Process	powershell.exe
Parent Process	powershell.exe
User	socadmin
Integrity Level	High
Suspicious Argument	-ep bypass

Investigation Analysis

Sysmon Event ID 1 captured PowerShell execution with execution policy bypass enabled.

The use of the `-ep bypass` argument significantly increased suspicion because attackers commonly use execution policy bypass techniques to disable PowerShell script restrictions and execute unauthorized scripts.

The nested PowerShell parent-child relationship further strengthened behavioral suspicion.

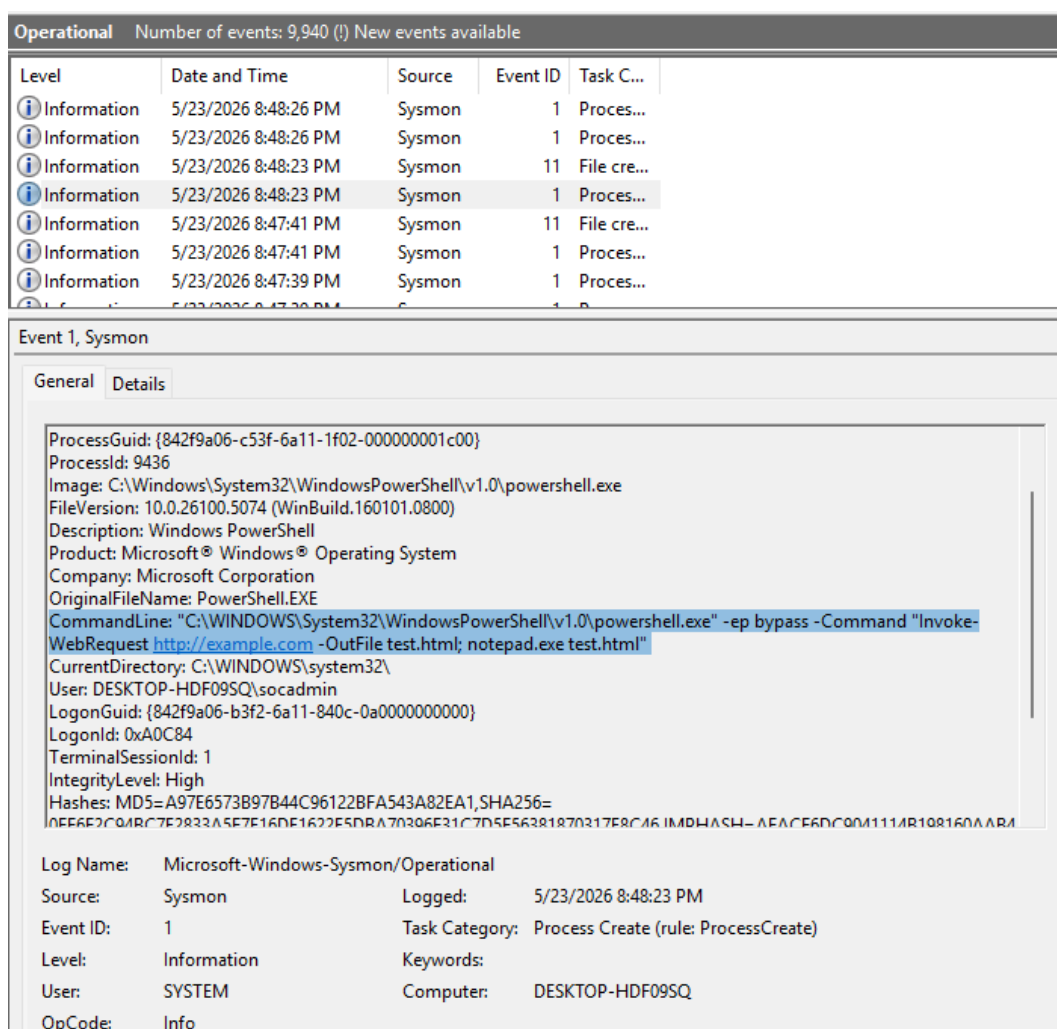


Figure 2 — Sysmon Event ID 1 showing PowerShell execution policy bypass activity

Sysmon Event ID 22 — DNS Query Analysis

Important Findings

Field	Value
Event ID	22
Process	powershell.exe
QueryName	example.com
QueryStatus	Success

Investigation Analysis

Sysmon Event ID 22 captured PowerShell initiating a DNS query for `example.com` prior to outbound network communication.

PowerShell-generated DNS activity is considered significantly more suspicious than standard browser DNS traffic because PowerShell is frequently abused for malware staging, payload retrieval, and command-and-control communication.

Operational

Number of events: 9,940 (!) New events available

Level	Date and Time	Source	Event ID	Task C...
Information	5/23/2026 8:48:41 PM	Sysmon	1	Proces...
Information	5/23/2026 8:48:27 PM	Sysmon	3	Networ...
Information	5/23/2026 8:48:27 PM	Sysmon	22	Dns qu...
Information	5/23/2026 8:48:26 PM	Sysmon	1	Proces...
Information	5/23/2026 8:48:26 PM	Sysmon	1	Proces...
Information	5/23/2026 8:48:23 PM	Sysmon	11	File cre...
Information	5/23/2026 8:48:23 PM	Sysmon	1	Proces...

Event 22, Sysmon

General

Details

Dns query:

RuleName: -

UtcTime: 2026-05-23 15:18:23.616

ProcessGuid: {842f9a06-c53f-6a11-1f02-000000001c00}

ProcessId: 9436

QueryName: example.com

QueryStatus: 0

QueryResults: 2606:4700:10::6814:179a;2606:4700:10::ac42:93f3;;;ffff:172.66.147.243;;;ffff:104.20.23.154;

Image: <unknown process>

User: DESKTOP-HDF09SQ\socadmin

Log Name: Microsoft-Windows-Sysmon/Operational

Source: Sysmon

Logged: 5/23/2026 8:48:27 PM

Event ID: 22

Task Category: Dns query (rule: DnsQuery)

Level: Information

Keywords:

User: SYSTEM

Computer: DESKTOP-HDF09SQ

OpCode: Info

Figure 3 — Sysmon Event ID 22 showing PowerShell DNS query activity

Sysmon Event ID 3 — Network Connection Analysis

Key Findings

Field	Value
Event ID	3
Process	powershell.exe
Destination Port	80
Protocol	HTTP
Initiated	true

Investigation Analysis

Sysmon Event ID 3 captured an outbound HTTP network connection initiated by PowerShell following DNS resolution activity.

The combination of PowerShell execution, DNS resolution, and outbound HTTP communication closely resembles behavioral patterns commonly observed in malware downloaders and staged payload delivery activity.

Analyst Note

Although the destination domain used during testing was benign, the behavioral sequence itself remained highly suspicious and demonstrated realistic attacker tradecraft.

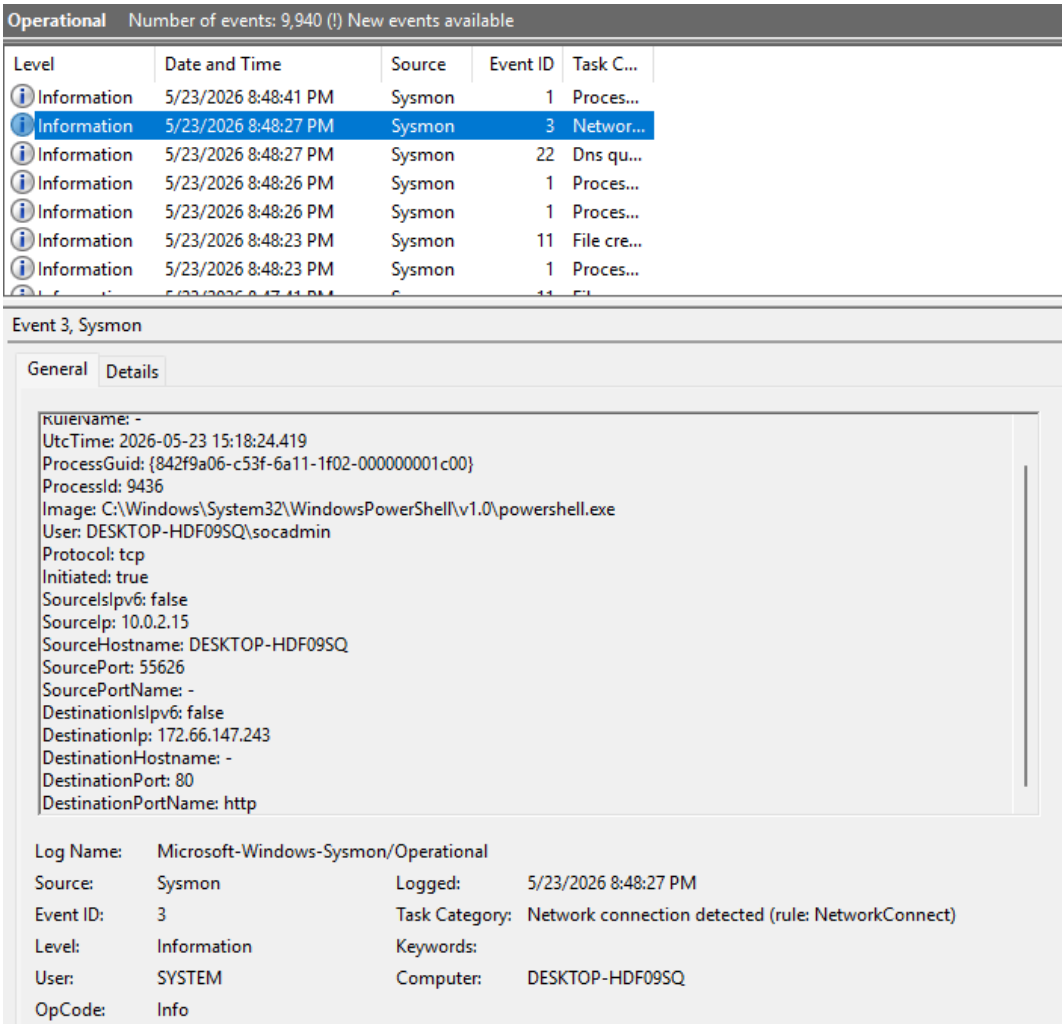


Figure 4 — Sysmon Event ID 3 showing outbound HTTP connection initiated by PowerShell

File Access and Process Chain Analysis

Downloaded File Investigation

Key Findings

Artifact	Value
Downloaded File	test.html

Artifact	Value
Parent Process	powershell.exe
Opened By	notepad.exe

Investigation Analysis

Following the outbound HTTP connection, the downloaded file was opened using Notepad through a PowerShell parent-child process relationship.

While the downloaded content itself was benign, the process chain demonstrated realistic post-download execution behavior commonly observed during malware staging and payload inspection activity.

Behavioral Correlation Analysis

The strongest detection indicator during this investigation was not a single event, but the behavioral correlation across multiple telemetry sources.

The complete attack chain consisted of:

- PowerShell execution
- execution policy bypass
- DNS query activity
- outbound HTTP communication
- file download behavior
- PowerShell child process creation

This significantly increased overall detection confidence.

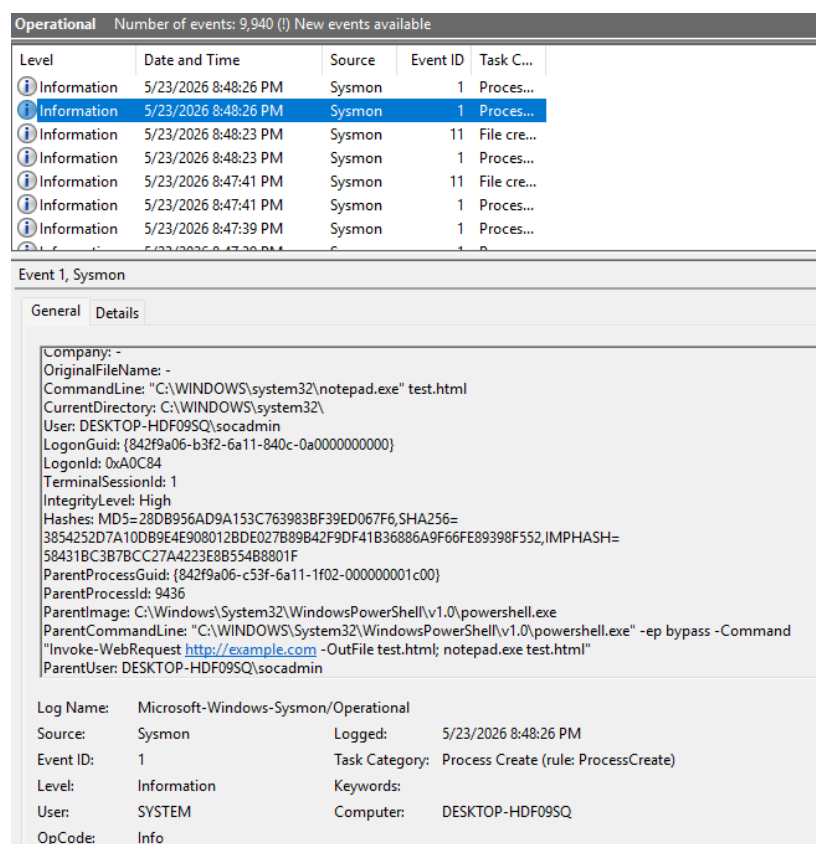


Figure 5 — Parent-child process relationship between PowerShell and Notepad

Wazuh Detection Analysis

Wazuh Rules Triggered

Rule ID	Severity	Description
Custom DNS Rule	Medium-High	PowerShell-generated DNS activity
Sysmon Correlation Rule	Medium-High	Suspicious PowerShell network behavior

Investigation Analysis

Wazuh successfully identified suspicious PowerShell-generated network activity through correlated Sysmon telemetry.

The investigation demonstrated how DNS queries and outbound network connections initiated by PowerShell can provide high-value behavioral indicators during malware staging investigations.

False Positive Validation

Although the destination domain and downloaded file used during testing were benign, the behavioral pattern itself remained highly suspicious due to the combination of PowerShell execution, network communication, and file interaction activity.

This highlights the importance of behavioral analysis over reputation-based detection alone.

```

> May 23, 2026 @ 20:48:24.186 input.type: log agent.ip: 192.168.56.106 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.image: C:\\WINDOWS\\System32\\WindowsPowerShell\\v1.0\\powershell.exe data.win.eventdata.processGuid: {84
2f9a06-c53f-6a11-1f02-000000001c00} data.win.eventdata.processId: 9436 data.win.eventdata.utcTime: 2026-05-23 15:18:23.845
data.win.eventdata.targetFilename: C:\\Users\\socadmin\\AppData\\Local\\Temp\\__PSScriptPolicyTest_lxxvi0aa.fje.ps1
data.win.eventdata.creationUtcTime: 2026-05-23 15:18:23.845 data.win.eventdata.user: DESKTOP-HDF09SQ\\socadmin

> May 23, 2026 @ 20:48:24.145 input.type: log agent.ip: 192.168.56.106 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.originalFileName: PowerShell.EXE data.win.eventdata.image: C:\\Windows\\System32\\WindowsPowerShell\\v1.0
\\powershell.exe data.win.eventdata.product: Microsoft Windows Operating System data.win.eventdata.parentProcessGuid: {842
f9a06-c515-6a11-1d02-000000001c00} data.win.eventdata.description: Windows PowerShell data.win.eventdata.logonGuid: {842f9a0
6-b3f2-6a11-840c-0a0000000000} data.win.eventdata.parentCommandLine: \"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powers

```

Figure 6 — Wazuh detection for suspicious PowerShell downloader activity

```

{
  "_index": "wazuh-alerts-4.x-2026.05.23",
  "_id": "tRpqVZ4Bdl3fDzgGxZ",
  "_version": 1,
  "_score": null,
  "_source": {
    "input": {
      "type": "log"
    },
    "agent": {
      "ip": "192.168.56.106",
      "name": "DESKTOP-HDF09SQ",
      "id": "001"
    },
    "manager": {
      "name": "wazuh-server"
    },
    "data": {
      "win": {
        "eventdata": {
          "image": "C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
          "processGuid": "{842f9a06-c53f-6a11-1f02-000000001c00}",
          "processId": "9436",
          "utcTime": "2026-05-23 15:18:23.845",
          "targetFilename":
            "C:\\\\Users\\\\socadmin\\\\AppData\\\\Local\\\\Temp\\\\__PSScriptPolicyTest_lxxvi0aa.fje.ps1",
          "creationUtcTime": "2026-05-23 15:18:23.845",
          "user": "DESKTOP-HDF09SQ\\\\socadmin"
        },
        "system": {
          "eventId": "11",
          "keywords": "0x8000000000000000",
          "providerGuid": "{5770385f-c22a-43e0-bf4c-06f5698ffbd9}",
          "level": "4",
          "channel": "Microsoft-Windows-Sysmon/Operational",
          "opcode": "0",
          "message": "\"File created:\\r\\nRuleName: -\\r\\nUtcTime: 2026-05-23 15:18:23.845\\r\\nProcessGuid:
            {842f9a06-c53f-6a11-1f02-000000001c00}\\r\\nProcessId: 9436\\r\\nImage:
            C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\r\\nTargetFilename:
            C:\\\\Users\\\\socadmin\\\\AppData\\\\Local\\\\Temp\\\\__PSScriptPolicyTest_lxxvi0aa.fje.ps1\\r\\nCreationUtcTime:
            2026-05-23 15:18:23.845\\r\\nUser: DESKTOP-HDF09SQ\\\\socadmin\"",

```



```

"version": "2",
"systemTime": "2026-05-23T15:18:23.8467902Z",
"eventRecordID": "15226",
"threadID": "4376",
"computer": "DESKTOP-HDF09SQ",
"task": "11",
"processID": "3596",
"severityValue": "INFORMATION",
"providerName": "Microsoft-Windows-Sysmon"
}
},
"rule": {
"firedtimes": 8,
"mail": true,
"level": 15,
"description": "Executable file dropped in folder commonly used by malware",
"groups": [
"sysmon",
"sysmon_eid11_detections",
"windows"
],
"mitre": {
"technique": [
"Ingress Tool Transfer"
],
"id": [
"T1105"
],
"tactic": [
"Command and Control"
]
},
"id": "92213"
},
"location": "EventChannel",
"decoder": {
"name": "windows_eventchannel"
},
"id": "1779549504.3258855",
"timestamp": "2026-05-23T15:18:24.186+0000"
},
"fields": {
"timestamp": [
"2026-05-23T15:18:24.186Z"
]
},
"sort": [
1779549504186

```

```
]
}
```

Figure 7 — Detailed Wazuh telemetry for PowerShell downloader behavior

Attack Timeline

Time	Event
20:48:23	PowerShell executed
20:48:27	DNS query generated
20:48:27	Outbound HTTP connection established
20:48:27	File downloaded
20:48:28	Notepad opened downloaded file

Key Detection Findings

Finding	Impact
PowerShell execution policy bypass	Defense evasion indicator
PowerShell DNS query activity	Suspicious network behavior
Outbound HTTP communication	Potential malware staging
File download activity	Payload retrieval simulation
Parent-child process chain	Behavioral correlation improvement

Primary Investigation Artifacts

Sysmon Event IDs

Event ID	Meaning
1	Process Creation
3	Network Connection
11	File Creation
22	DNS Query

Wazuh Detection IDs

Rule ID	Severity	Description
92027	4	PowerShell spawned nested PowerShell instance
100100	10	PowerShell initiated DNS query
92213	15	Suspicious file creation in Temp directory

Investigation Outcome

The investigation successfully demonstrated how PowerShell downloader activity can be detected using behavioral telemetry collected from Sysmon and correlated through Wazuh detection logic.

The combination of:

- PowerShell execution analysis
- DNS telemetry
- network connection monitoring
- file interaction visibility
- parent-child process correlation

significantly improved detection confidence and investigation quality.

The investigation also reinforced the importance of:

- behavioral correlation
 - contextual analysis
 - false-positive validation
 - PowerShell monitoring
 - structured SOC investigation methodology
-

Conclusion

This simulation demonstrated how Sysmon and Wazuh provide strong visibility into PowerShell-based downloader activity involving DNS queries, outbound HTTP communication, and suspicious network-enabled PowerShell execution.

Behavioral correlation between PowerShell execution, DNS telemetry, network connections, and file interaction activity significantly improved detection confidence and investigation quality.

The investigation also highlighted the importance of multi-event behavioral analysis when identifying suspicious PowerShell activity commonly associated with malware staging and payload delivery.